

GlucoTwin

A diabetes digital twin that predicts glucose spikes 30 minutes to 2 hours before they happen

Digital Twin Challenge 2026 · Team hackers_ · IIT (ISM) Dhanbad · Team leader: Korra Santhosh

Diabetes care in India is still reactive

101 M

Indians living with diabetes

136 M

more with prediabetes (ICMR-INDIAB, 2023)

- Repeated high glucose (> 180 mg/dL) drives eye, kidney, nerve and heart damage.
- A continuous glucose monitor alarms only after glucose is already high.
- Doctors see past readings, not where this patient is heading next.



Our question, asked every 15 minutes for every patient:

Will glucose go above 180 mg/dL in the next 30 minutes, 1 hour or 2 hours?

Our digital twin fuses two data streams and looks ahead



Static stream: EHR

Age, BMI, diabetes type and duration, complications, medicines, HbA1c and other labs



Dynamic stream

CGM glucose, insulin, meals, heart rate, steps — every 5 to 15 minutes



The twin

111 fused features → three models (30 min, 1 h, 2 h), each with accuracy and safety alert modes



Doctor dashboard

Live timeline, risk cards, patient record and a replay of what really happened

Built only on open, anonymised data: 187 people, Type 1 and Type 2

Dataset	People	Diabetes	What it adds
ShanghaiT2DM	100	Type 2	Real clinical record (labs, drugs, history)
ShanghaiT1DM	12	Type 1	Real clinical record
HUPA-UCM (Spain)	24	Type 1	Fitbit heart rate, steps, calories
AZT1D (USA)	25	Type 1	Insulin-pump data, carbohydrates
T1D-UOM (UK)	17	Type 1	Insulin, meals, heart rate, steps
BrisT1D (UK)	9	Type 1	Insulin, meals, smartwatch
OhioT1DM (USA)	12	Type 1	External test only — never trained on

187

people for training & testing

≈ 530 k

prediction moments per horizon

Open data

CC BY 4.0 research datasets*

* BrisT1D is published under CC BY 4.0; we used its Kaggle competition copy, whose rules we have not separately verified.

We found and fixed six data problems before any modelling

Duplicated readings

BrisT1D stored each reading up to 72 times; stitched into one timeline (0 conflicts in 11.4 M values)

Hidden future leak

HUPA-UCM gaps were filled by straight lines toward the NEXT reading; removed

Mixed units

mmol/L vs mg/dL, carbs in grams vs 10-g portions, basal U/h vs units

Missing ≠ zero

"Not recorded" kept separate from "none"; long-acting insulin spread over 24 h

Different sensor speeds

5-min and 15-min CGMs put on one common 15-minute grid

Leak-proof by test

100+ automated tests: changing future data never changes a past feature

111 fused features feed three horizon-specific models (123 in the v9 2-hour model)

Stream features · 80

- Glucose level, change, slope, acceleration, 30-min projection
- Rolling mean, spread, min, max (30 min – 4 h)
- Insulin on board, carbs on board, last meal
- Heart rate, steps, energy
- Personal memory: spikes at this time on the last 7 days

EHR features · 31

- Age, sex, BMI, smoking, alcohol
- Diabetes type and duration, complications
- HbA1c, glycated albumin, fasting and post-meal glucose, C-peptide
- Lipids, creatinine, eGFR
- Drug classes: insulin, metformin, SGLT2, DPP-4, GLP-1 …



HistGradientBoosting × 3

- One model per horizon: 30 min, 1 h, 2 h
- Accuracy mode and safety mode thresholds
- Each person weighted equally
- 2-hour model v9: +12 history and trend features
- Runs live in under a second

Every number is measured on people the model never saw

1	**Person-grouped 7-fold CV**	A person is never in training and testing at once
2	**Nested thresholds**	Alert thresholds chosen inside the training folds only
3	**Leave-one-dataset-out**	Train on 4 datasets, test on a 5th
4	**External test: OhioT1DM**	New hospital, devices and patients
5	**Bias & fit audit**	Sex, age, BMI, diabetes type, time of day, calibration, train-vs-test gap

Early warning that works: 94.7% at 30 minutes, 84.9% at 2 hours

Next 30 min

94.7%

accuracy on unseen people

ROC-AUC 0.983

Spikes caught 86.4% (safety mode 92.3%)

OhioT1DM external: 94.5%

Next 1 hour

91.1%

accuracy on unseen people

ROC-AUC 0.956

Spikes caught 77.8% (safety mode 85.3%)

OhioT1DM external: 90.6%

Next 2 hours

84.9%

accuracy on unseen people

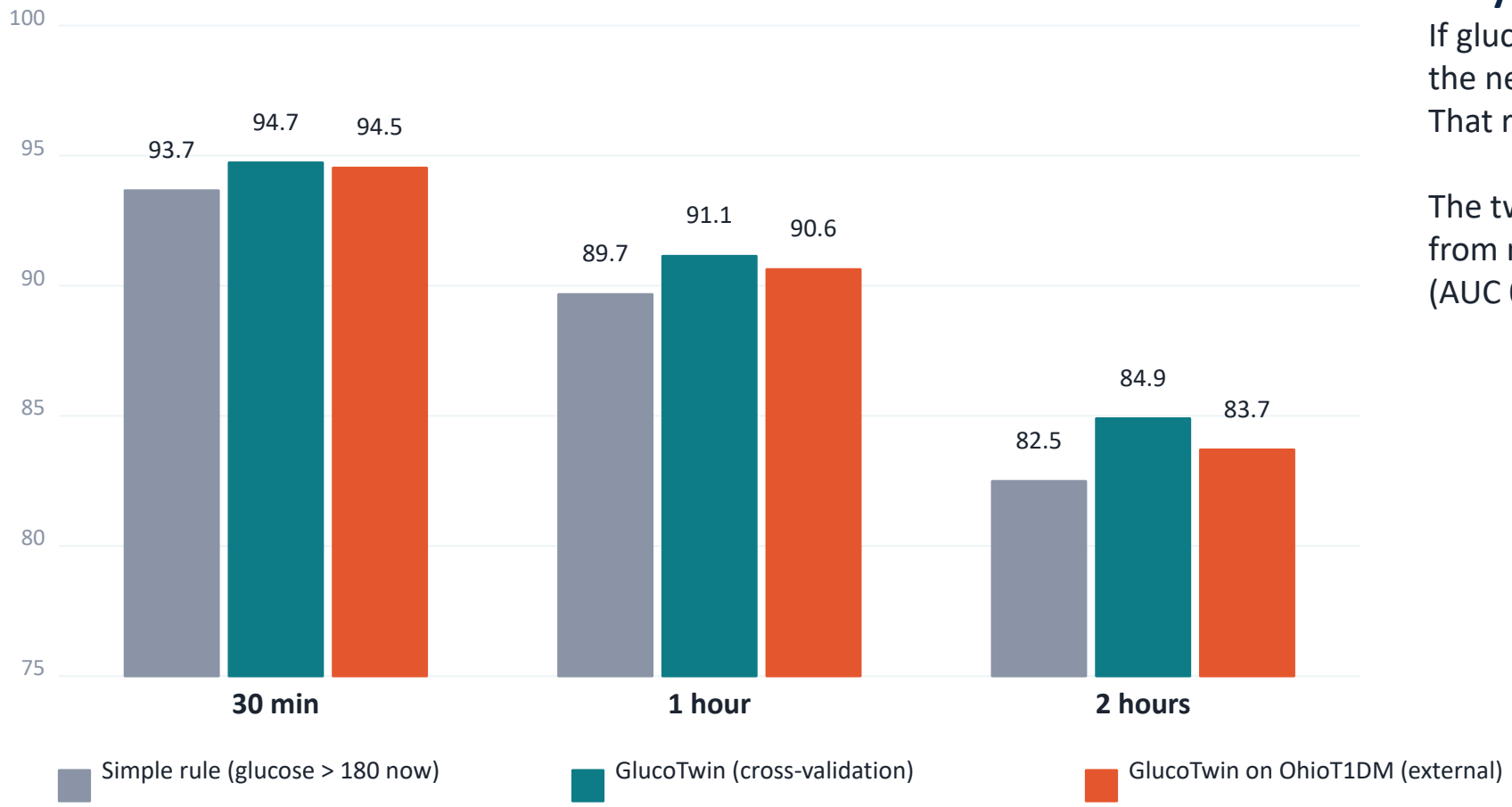
ROC-AUC 0.913

Spikes caught 68.9% (safety mode 76.7%)

OhioT1DM external: 83.7%

GlucoTwin beats the "already high" rule, and the margin grows with the horizon

Accuracy on people the model never saw (%)



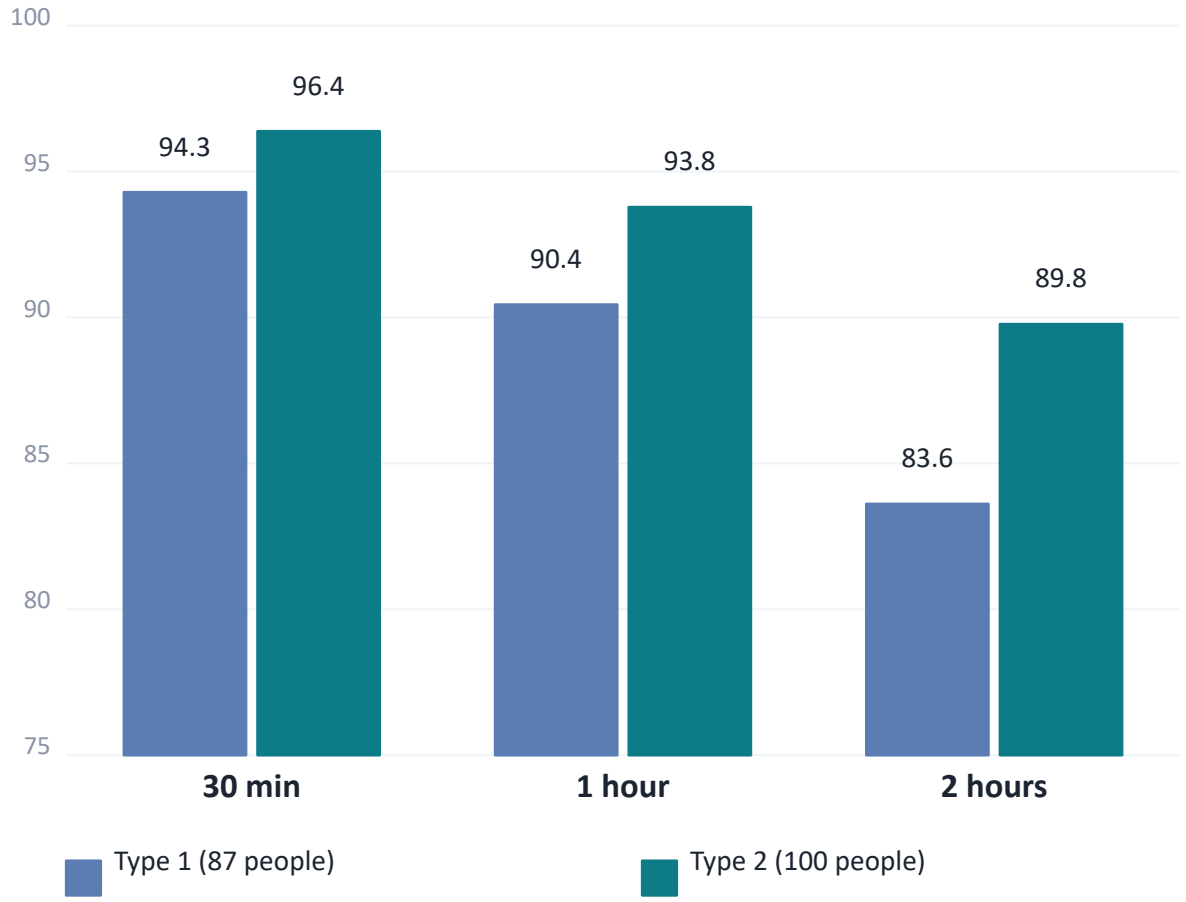
Why a baseline matters

If glucose is already above 180, a spike "in the next 30 minutes" is almost certain. That rule alone is right most of the time.

The twin adds the hard part: spotting rises from normal glucose. Its ranking quality (AUC 0.91 at 2 h) is far above the rule's.

Strongest for Type 2 diabetes, and fair across patient groups

Accuracy by diabetes type (%)



Group (Type 2 & Type 1, Shanghai)	2-h accuracy	Calibration gap
Women	90.0%	0.2 pts
Men	89.5%	-0.8 pts
Age < 40	92.5%	0.2 pts
Age 40–60	89.4%	-0.7 pts
Age 60+	89.4%	-0.2 pts
BMI < 23	89.4%	-0.3 pts
BMI 23–27.5	89.1%	-0.6 pts
BMI 27.5+	94.8%	0.9 pts

What we learned by testing every idea

Helped

- More diverse patients: 21 → 187 people from 4 countries
- Personal memory: "spikes at this time in the last 7 days"
- Cleaning the data (removing a future leak, fixing units)
- Shorter horizons for urgent alerts; safety-mode thresholds
- Type 2 data: the most accurate group (89.8% at 2 h)

Did not help (measured, not guessed)

- Bigger / deeper models: they only memorise training patients*
- Even perfect knowledge of future meals: +0.2 points*
- Per-patient recalibration on day 1–3: no gain*
- EHR on top of CGM: +0.2 points in a patient's first day (not significant), ~0 after
- Extra wearable features (steps, calories): +0.0 points*

* measured in the earlier 87-person study (v6); the other results are from the 187-person v7 evaluation

Where the remaining error comes from

49.9%

of 2-hour errors happen where the actual peak was 162–198 mg/dL, i.e. within ±10 % of 180

Accuracy 90.2% on clear outcomes vs 66.8% near 180 (530,807 moments, 187 people)

13.9%

mean difference between CGM and fingerstick readings (MARD): the reference itself is noisy

4,010 pairs from 98 Shanghai patients; median 11.0%; CGM reads 7.0% low

36.8%

of serious spike episodes get no warning 30–120 min ahead

3,495 of 9,503 episodes (peak ≥ 250 or ≥ 3 consecutive 15-min bins > 180)

Calibration on held-out people is good (ECE 0.15 / 0.34 / 0.38 % at 30 min / 1 h / 2 h). "Near 180" uses the actual outcome, so it splits the error rather than selecting cases in advance. Fingerstick meters have their own error.

New information, tested once on 37 people locked away in advance

Prediction (2 h)

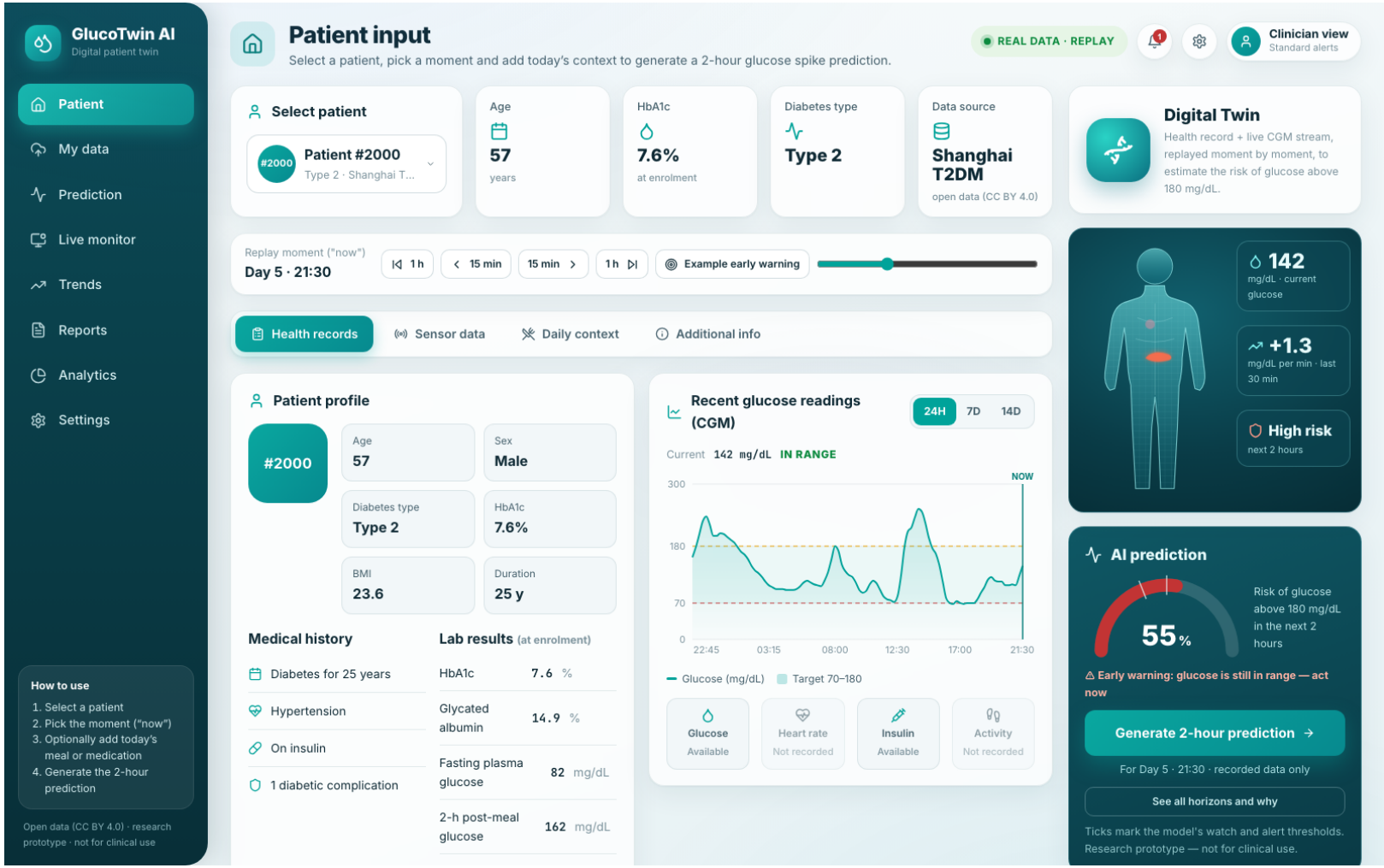
- Accuracy 85.24% → 85.33% (change within noise)
- Ranking (AUC) 0.9154 → 0.9167: +0.0013, significant
- OhioT1DM external 83.72% → 83.98%
- Type 2 90.41% → 90.37%

Alerts (serious spikes)

- Warned 30–120 min ahead: 1,526 → 1,569 of 2,515 episodes
- False alerts per patient-day 1.46 → 1.33
- All alerts per day 8.6 → 3.7: warns only while glucose is still in range

Tested: longer personal history, recorded meals and tablets, multi-window trends (all known at prediction time). Every choice was made on 150 development people; the 37 test people were scored once. Served: v9 2-hour model + early-warning policy; 30 min / 1 h unchanged. No open Indian Type 2 CGM dataset was available.

What the doctor sees



Patient input
EHR, labs, CGM chart, data completeness

Digital twin & AI prediction
Live probability from FastAPI, early warning

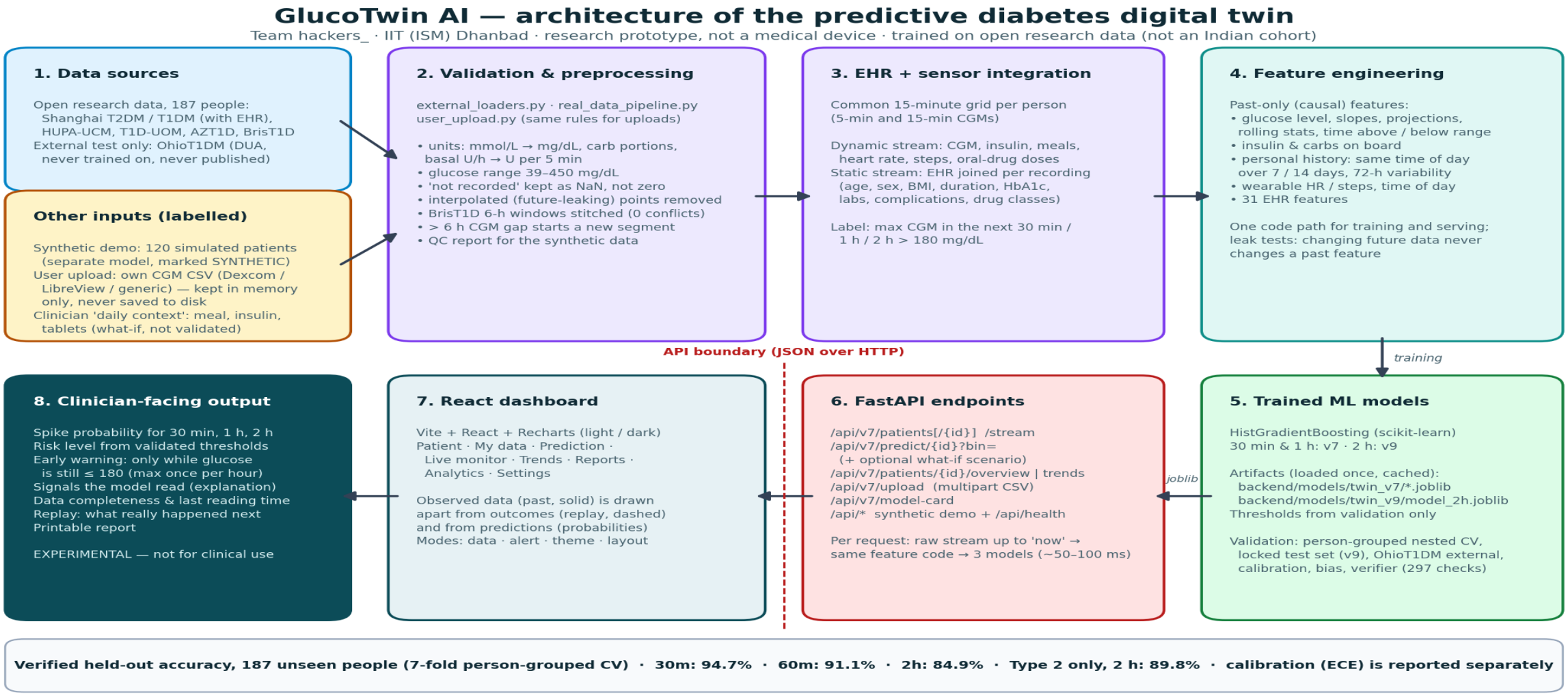
Daily context (what-if)
Meal, insulin or tablets taken now

Monitor · Trends · Reports · My data
Replay, time in range, printable report, upload your own CGM file

Demo workflow: from patient to prediction in five steps

1	**Select a patient**	30 open-data demo patients (Type 1 and Type 2) or your own uploaded CGM file
2	**Pick the moment**	Replay slider sets "now"; the model only sees data up to that moment
3	**Review the record**	EHR and labs, CGM history, sensor completeness
4	**Add daily context**	Optional what-if: meal carbs, rapid insulin, oral tablets
5	**Generate & compare**	Probabilities for 30 min, 1 h, 2 h; then the replay shows what really happened

Architecture



Limitations and clinical safety

Not clinically validated

A research prototype trained on open research data; not a medical device and not for treatment decisions.

Not an Indian cohort

The Type 2 data is from Shanghai, China. Validation with Indian Type 2 patients is future work.

Missed serious spikes

36.8% of 9,503 serious episodes got no warning 30–120 min ahead (all 187 people, event level).

Errors near the 180 line

49.9% of 2-hour errors have a peak within ±10% of 180 mg/dL; CGM and fingerstick differ by MARD 13.9%.

False alarms

Even with the in-range early-warning rule, about 1.3 false alerts per patient-day on the locked test set.

What-if and uploads

Daily-context scenarios and individual uploads are not separately validated.

From prototype to Indian clinics



Clinical validation

Prospective study with Indian Type 2 patients and clinicians before any real use



Health-record integration

Connect to ABHA-linked records (ABDM) for labs and prescriptions



Affordable sensing

Low-cost CGMs and phone alerts for patients and care teams



Privacy by design

DPDP-Act consent, encrypted processing, clinician-in-the-loop alerts



See the spike before it happens

94.7% at 30 min · 91.1% at 1 h · 84.9% at 2 h — on 187 unseen people, confirmed on an external dataset

hackers_ · IIT (ISM) Dhanbad · Code (MIT): github.com/KorraSanthosh/happiness_ · Thank you